

7	(a)	The <u>magnitude</u> of the <u>induced e.m.f.</u> is <u>directly proportional</u> to the <u>rate of change of magnetic flux linkage</u> .	B1
	(b)(i)	Final flux linkage = $8.0 \times 10^{-3} \times 0.4 \times 10^{-4} \times 150$ $= 4.8 \times 10^{-5} \text{ Wb}$ Initial flux linkage = $50 \times 10^{-3} \times 0.4 \times 10^{-4} \times 150$ $= 3.0 \times 10^{-4} \text{ Wb}$ change = $(4.8 \times 10^{-5}) - (3.0 \times 10^{-4})$ $= 2.52 \times 10^{-4} \text{ Wb}$	M1 A1
	(ii)	Ave e.m.f. = $\frac{\Delta\phi}{\Delta t}$ $= \frac{2.52 \times 10^{-4}}{0.30}$ $= 8.4 \times 10^{-4} \text{ V}$	M1 A1
	(c)	The <u>change in flux linkage over a given distance</u> <u>decreases</u> when <u>the coil is further</u> from the magnet.	M1 M1

6

		For the induced emf to remain the same, the <u>rate of change of flux linkage with respect to time must be constant.</u> Thus, the <u>speed</u> of the coil should <u>increase</u> .	A1
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8	(c) (i)	In photoelectric effect, EM radiation releases electrons	
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